A Two-Thirds Hyperbolicity Wedge for Jensen Polynomials of Riemann's Xi-Function

Reader's synopsis and proof map

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Purpose and review status. This is a compact synopsis; the maintained scholarly sources are the main paper and technical supplement in the `paper` directory. The user-executed exact Mathematica reconstruction completed the independent-CAS checks M1–M4. Lean now checks the complete T1–T5 analytic chain, the exact xi branch through six matches, and the critical-radius conclusion under narrow typed literature inputs. The chosen Phase-30 endpoint also constructs the final xi-specific sixth-multiplier certificate, connects it to the actual transformed Jensen polynomial, and checks the headline conclusion conditional only on those typed inputs. All available reviews are AI-assisted evidence, not human or peer review.

Abstract

Let $\gamma(n)$ be the Taylor coefficients of Riemann's completed ξ-function in the standard even normalization, and let

$$\mathcal{J}^{d,n}(X) = \sum_{j=0}^{d} \binom{d}{j} \gamma(n+j) X^j.$$

We prove that there is an absolute $K > 0$ such that $n^2 \log(n+2) \geq K d^3$ implies that $\mathcal{J}^{d,n}$ has d distinct negative real zeros. The gain from the earlier three-fifths architecture comes from matching six logarithmic coefficients with a four-parameter product of two Jacobi factors. The proof combines a direct sectorial saddle analysis for the xi coefficients, a sixth-order residual estimate, finite-free convolution, and a non-perturbative critical-point recurrence. We state the trust boundary precisely: decisive symbolic identities have exact SymPy/Lean regression evidence and an exact user-executed Mathematica second-CAS reconstruction, and the complete T1–T5 analytic chain is kernel-checked in Lean. The xi-specific multiplier, its six nodes and unit bound, interval certificate, and headline assembly are also kernel checked. The remaining formal boundary consists only of typed Jacobi, MMP, and MSS literature inputs. This synopsis compresses that proof; detailed inequalities and evidence maps are in the main paper and supplement.

1 Statement and architecture

Theorem 1.1 (Two-thirds wedge)**.** *There is an absolute constant $K > 0$ such that for integers $d \geq 1$ and $n \geq 0$, the implication*

$$n^2 \log(n+2) \geq K d^3$$

ensures that $\mathcal{J}^{d,n}$ has exactly d distinct negative real zeros.

The proof is developed block by block below and is closed explicitly in Section 11. The synopsis uses ordinary section-based theorem numbers; the labels T1–T18 in the expanded paper and assurance matrix are stable evidence identifiers rather than additional theorem numbers. Thus the sectorial coefficient theorem is T5 and the present main theorem is T18.

The proof is effective in principle after tracing the saddle and branch constants, but the present unweighted sup-norm contraction yields an astronomically large sufficient threshold. We make no claim that the resulting constant is computationally useful, and we do not print a numerical value of K.

The proof has four blocks. First, Sections 2–5 derive a sectorial coefficient asymptotic and its logarithmic derivatives through order six. Second, Sections 6–7 choose a positive four-parameter comparison model matching six coefficients. Third, Sections 8–9 establish positive, simple, localized roots and the derivative radius of that comparison. Finally, Sections 10–11 transfer signs to the actual Jensen polynomial.

Throughout, a statement called "uniform" is uniform after the fixed nested sectors and compact parameter box have been chosen. Holland's main theorem is not a premise. His paper supplies architecture and several formulas which are rederived or connected directly to their primary sources.

2 The Mellin coefficient identity

Put

$$\Theta(t) = \sum_{m\geq 1} e^{-\pi m^2 t}, \qquad \omega(t) = \tfrac{1}{2}\big(2t^2\Theta''(t) + 3t\Theta'(t)\big).$$

Riemann's integral and the substitution $t = e^{2u}$ give an even Fourier– Laplace representation of Ξ. Differentiating at the origin, retaining both halves of the even integral, gives the exact normalization

$$\gamma(n) = \frac{8\,n!}{(2n)!}\int_0^\infty \omega(e^{2u})e^{u/2}u^{2n}\,\mathrm{d}u. \tag{1}$$

The factor eight in (1) is important when the main term is compared character for character with the printed GORZ/GORTTW convention. It cancels from logarithmic derivatives but not from a source-fidelity check. Indeed Riemann's kernel in these variables is $8e^{u/2}\omega(e^{2u})$, and extracting the coefficient of w^{2n} from $\cosh(wu)$ contributes $1/(2n)!$. Since our convention is $\xi(1/2+w) = \sum_{n\geq 0}\gamma(n)w^{2n}/n!$, this gives (1). Equivalently, the prefactor is $8\sqrt{\pi}/(4^n\Gamma(n+1/2))$ by duplication. No asymptotic input enters.

Expanding Θ separates the $m = 1$ mode from $m \geq 2$. If M denotes the complex Mellin exponent, the leading phase has one saddle and the remaining theta modes acquire an additional negative exponential proportional to $m^2 - 1$. This exact decomposition is the starting point for the sectorial analysis.

3 The sectorial saddle variable

For $0 < \theta < \pi/2$, use principal logarithms in $|\arg N| \leq \theta$, and set

$$\Psi(L) = L\left(\pi e^L + \frac{3}{4}\right), \quad w = \log N, \quad L_0 = w - \log w - \log\pi.$$

Lemma 3.1 (Sectorial saddle, Lemma S)**.** *For $|N| \geq e^{12}$ and $|\arg N| \leq \theta$ there is a unique holomorphic branch L_N, agreeing with the positive real solution, such that $N = \Psi(L_N)$ and $|L_N - L_0| < 1$. Uniformly on every fixed closed smaller sector,*

$$|L_N| \asymp \log|N|, \qquad r := L_N^{-1} \longrightarrow 0, \qquad \sigma := L_N/N \longrightarrow 0.$$

Moreover $|L_N| \geq \frac{1}{2}\log|N|$, $|r|, |\sigma| \leq 7/50$, and

$$Q_N = (1 + L_N)N - \tfrac{3}{4}L_N^2 = NL_N(1 + r - \tfrac{3}{4}\sigma), \tag{2}$$

$$|1 + r - \tfrac{3}{4}\sigma| \geq 9/16, \qquad L_N' = L_N/Q_N. \tag{3}$$

Proof. Consider

$$G(L) = L + \log L + \log\pi - \log\left(1 - \frac{3L}{4N}\right) - \log N$$

on $|L - L_0| \leq 1$. If $\ell = \log|N| \geq 12$, define

$$m(\ell) = \ell - \log(\ell + 1) - \log\pi - 1, \qquad M_\theta(\ell) = \ell + 2 + \log(\ell + 1) + \theta/\ell + \log\pi.$$

Then

$$\Re L \geq m(\ell) > 0$$

and

$$|L| \leq M_\theta(\ell).$$

Thus all principal logarithms are legal and $|3L/(4N)| < 10^{-4}$. On the boundary, compare G with $H(L) = L - L_0$. The elementary estimate $|\log(1 + z)| \leq 2|z|$ for $|z| \leq 1/2$ gives

$$\sup_{\partial D}|G - H| \leq 2\frac{\log(\ell + 1) + \theta/\ell + \log\pi + 1}{\ell} + \frac{3M_\theta(\ell)}{2e^\ell} < 1.$$

Rouche's theorem gives one simple zero. Exponentiating at that zero yields $N = \Psi(L)$. Simplicity and the holomorphic implicit-function theorem patch the local zeros into a branch; conjugation and real uniqueness identify the positive real branch. The displayed disc bounds give the asserted asymptotics. In particular $m(\ell) > \ell/2$ for $\ell \geq 12$, so $|L_N| \geq \Re L_N \geq \ell/2$.

For (3), the reverse triangle inequality and $|r|, |\sigma| \leq 7/50$ give

$$|1 + r - \tfrac{3}{4}\sigma| \geq 1 - \frac{7}{50} - \frac{3}{4}\frac{7}{50} = \frac{151}{200} > \frac{9}{16}.$$

Hence $Q_N \neq 0$ before implicit differentiation is invoked. Differentiating $N = \Psi(L_N)$ then gives $L_N' = L_N/Q_N$. □

The same exact argument gives

$$|4 + 4r - 3\sigma| \geq 4 - 4\frac{7}{50} - 3\frac{7}{50} = \frac{151}{50}. \tag{4}$$

This is the reduced denominator margin used at derivative order six.

4 Contour localization and the coefficient theorem

Theorem 4.1 (Sectorial coefficient asymptotic)**.** *Fix $\theta_0 = 1/400$ and $\theta_1 = 1/200$. On the inner sector, the analytic continuation of the xi coefficient moment equals its explicit saddle main term times*

$$1 + O\left(\frac{\log|M|}{|M|}\right),$$

uniformly as $|M| \to \infty$. The main term is the one obtained from (1) *and agrees, after the factor-eight normalization, with the printed GORZ main term. It is nonzero on the inner sector.*

Proof. The leading $m = 1$ Mellin integral is moved, in $u = \log t$, to the horizontal ray through the saddle supplied by Lemma 3.1. The move stays in the principal-log domain. For $|\arg s| \le 1/100$, the logarithmic saddle equation and $\Re L_s \ge m(12) > 7.29$ give $|\Im L_s| < 1/20$. Writing

$$K_s = \frac{s}{L_s}\left(1 + \frac{1}{L_s} - \frac{3L_s}{4s}\right),$$

the corresponding argument estimate gives $|\arg K_s| < 1/20$ and $\Re K_s \ge \cos(1/20)|K_s|$. This right-half-plane bound—not merely the nonvanishing consequence of (3)—controls the Gaussian and the strict concavity on the shifted ray. The two endpoint connectors are $\mathcal{A}(s)O(e^{-c|s|\log\log|s|})$, where $\mathcal{A}(s)$ is the saddle main term. Splitting into a central Gaussian window and two tails gives a relative error $O(|K_s|^{-1})$; the signed cubic Gaussian moment removes the otherwise larger absolute error.

For $m \ge 2$, the same contour is legal and the extra theta exponent is uniformly negative. Summing the modes is therefore exponentially smaller than the first mode. Sectorial Stirling, applied on fixed nested sectors, then yields the asserted main term and relative error. The detailed connector, central-window, signed-cubic, tail, mode-summation, and Stirling estimates are expanded in the main paper and technical supplement.

The proof just described is carried out with fixed sector/radius constants; there is no assertion that every later error constant is uniform for an arbitrary free $\theta \in (0, \pi/2)$. ☐

5 Logarithmic derivatives through order six

Let $h(x)$ be the logarithm of the exact moment normalization on the positive axis and let $\mathcal{L}_x = L_{2x-2}$. Cauchy's inequality on proportional discs inside the two fixed sectors differentiates the relative error in Theorem 4.1. Total implicit differentiation of the explicit main term gives

$$\begin{aligned}
h''(x) &= \frac{2}{x\mathcal{L}_x}(1 + O(\mathcal{L}_x^{-1})),\\
h^{(3)}(x) &= -\frac{2}{x^2\mathcal{L}_x}(1 + O(\mathcal{L}_x^{-1})),\\
h^{(4)}(x) &= \frac{4}{x^3\mathcal{L}_x}(1 + O(\mathcal{L}_x^{-1})),\\
h^{(5)}(x) &= -\frac{12}{x^4\mathcal{L}_x}(1 + O(\mathcal{L}_x^{-1})),\\
h^{(6)}(x) &= \frac{48}{x^5\mathcal{L}_x}(1 + O(\mathcal{L}_x^{-1})).
\end{aligned} \tag{5}$$

Only the bound in the last line is needed. Before the chain rule $N = 2x - 2$, the exact sixth derivative has the form

$$G_0^{(6)}(N) = \frac{H_6(r,\sigma)}{N^5 L_N}, \qquad \operatorname{den} H_6 = (4 + 4r - 3\sigma)^{12}.$$

The numerator is a degree-thirteen polynomial with 82 terms. An exact coefficientwise majorant on $|r|, |\sigma| \le 7/50$, together with (4), is less than 10^4. This gives the conservative uniform bound $|G_0^{(6)}(N)| \le 20000/(|N|^5 \log|N|)$.

The rational differentiation tower and its coefficientwise majorant are machine-checked artifacts. We do not duplicate that decisive symbolic algebra here; Section 13 records its exact independent-CAS reconstruction and frozen evidence.

6 Six coefficient matches and the elementary map

Use positive parameters

$$A = \frac{\alpha}{xe}, \quad B = \frac{t + we}{x}, \quad C = \frac{t}{x}, \quad D = \frac{1 + \delta e}{x}, \qquad x = n^{-1},\ e = \mathcal{L}_n^{-1}.$$

The compact box is

$$\alpha \in [5/2, 7/2], \quad t \in [7/4, 9/4], \quad w \in [5, 6], \quad \delta \in [1/4, 5/12].$$

For $f_k(U) = \log(U + k) - \log(U + k + 1)$, the exact gamma half-shift is $-f_k(n + 1/2)$. Pairing that term before estimation avoids a spurious $\mathcal{L}_n$ loss. Repeated FTC gives, with $q = j + 1$,

$$\Delta^j f_0(U) = (-1)^{j+1} j! \int_{[0,1]^q} (U + u_1 + \cdots + u_q)^{-q}\, \mathrm{d}u.$$

Writing

$$\Phi_q(s, z) = \int_{[0,1]^q} (s + z \textstyle\sum u_i)^{-q}\, \mathrm{d}u,$$

one has, for $r = 0, 1, 2$,

$$\partial_s^r \Phi_q(s, z) = (-1)^r (q)_r \int_{[0,1]^q} (s + z \textstyle\sum u_i)^{-q-r}\, \mathrm{d}u.$$

If $s \ge s_0 > 0$ and $z \ge 0$, the unit cube has volume one and hence

$$|\partial_s^r \Phi_q(s, z)| \le (q)_r s_0^{-q-r}.$$

The mean-value theorem adds the explicit first-order error $(q)_r (q + r) q z s_0^{-q-r-1}$.

With $(a_0, a_1, a_2, a_3) = (1, 1/2, 1, 1)$, the normalized elementary term is

$$\begin{aligned} H_{n,j} = \frac{a_j}{e} \{ & e^q \Phi_q(\alpha, xe) - \Phi_q(t + we, x) + \Phi_q(t, x) \\ & - \Phi_q(1 + \delta e, x) + \Phi_q(1 + x/2, x) \}. \end{aligned}$$

The $B - C$ divided difference tends to qw/t^{q+1}; the paired D–gamma divided difference tends to $q\delta$; and the remote boundary contributes $1/\alpha$ only for $q = 1$. Values and first parameter derivatives converge uniformly, giving

$$H^\infty = \left(\frac{1}{\alpha} + \frac{w}{t^2} + \delta, \frac{w}{t^3} + \delta, \frac{3w}{t^4} + 3\delta, \frac{4w}{t^5} + 4\delta \right)$$

with error $O_K(e + x/e)$ in value and $O_K(e + x)$ in derivatives. Combining (5) with the exact scales gives the xi contribution $S^\infty = (-2, -1, -2, -2)$.

Lemma 6.1 (Quotient adapter)**.** *Let (a_j) and (b_j) be nonzero coefficients with compatible logarithm branches. Equality of the four consecutive quotient coordinates is equality of the four second differences*

$$(a_{k+2} - 2a_{k+1} + a_k) = (b_{k+2} - 2b_{k+1} + b_k), \quad 0 \le k \le 3,$$

in logarithmic coordinates. Together with the two normalizations at indices 0 *and* 1*, these equations imply equality at all six indices* $0, \dots, 5$*.*

Proof. For the difference $c_k = a_k - b_k$, the hypotheses say $c_0 = c_1 = 0$ and $c_{k+2} = 2c_{k+1} - c_k$. Induction gives $c_2 = \cdots = c_5 = 0$. □

7 The positive parameter branch

The limiting residual map $F = H^\infty + S^\infty$ has the positive zero

$$y_* = (\alpha, t, w, \delta) = (3, 2, 16/3, 1/3).$$

Exact elimination gives positive-orthant uniqueness. Its Jacobian satisfies

$$\det DF(y_*) = -\frac{1}{144}, \qquad \|DF(y_*)^{-1}\|_\infty = \frac{304}{3}.$$

On a smaller rational box K_0 around y_*, choose $P = DF(y_*)^{-1}$ so that $\|I - PDF\|_\infty \le 1/4$. The C^1 estimate in Section 6 makes $T_n(y) = y - PG_n(y)$ a self-map and a contraction with constant at most $1/2$. Thus there is a unique zero in K_0 and

$$y_n - y_* = O(\mathcal{L}_n^{-1}).$$

Uniqueness is asserted only in K_0, not globally.

For $0 < e \le 1/12$ and $x > 0$, exact endpoint arithmetic gives

$$\frac{\alpha}{e} \ge 30 > \frac{11}{4} \ge t + we, \qquad t - 1 - \delta e \ge \frac{103}{144} > 0.$$

Consequently $A > B > C > D > 0$ on the box. The point y_* has exact left/right margins $(1/2, 1/2)$, $(1/4, 1/4)$, $(1/3, 2/3)$, and $(1/12, 1/12)$.

8 Finite-free comparison and localization

The comparison polynomial is

$$p_F(y) = {}_3F_2\left(\begin{matrix} -d, A, C \\ B, D \end{matrix}; \frac{D}{AC} y\right).$$

In constant-term-one normalization it is the multiplicative finite-free convolution of two scaled Jacobi factors. Reversal converts this identity to the monic descending convention in the finite-free literature. To pin the convention, order positive roots as $\lambda_1 \ge \cdots \ge \lambda_d > 0$ and, following MMP Definition 2.16, put

$$\operatorname{lmesh}(p) = \min_{1 \le j < d} \frac{\lambda_j}{\lambda_{j+1}} \ge 1.$$

Each Jacobi factor has distinct positive roots, so its logarithmic mesh is strictly greater than one. MMP Proposition 2.7(iii) preserves nonnegative real-rootedness, while Proposition 2.17 states in this convention that

$$\operatorname{lmesh}(p \boxtimes_d q) \geq \operatorname{lmesh}(p).$$

Thus the convolution is simple. Since $p_F(0) = 1$, zero is not a root. The exact proposition text is pinned to the accessible arXiv v3; the paywalled journal PDF has not been byte-compared.

For a Jacobi factor $q_{U,V}$, direct Gershgorin estimates on the transported Jacobi matrix give the ratio-free statement

$$U \geq V + d, \quad V \geq 32d \quad \Longrightarrow \quad \operatorname{roots}(q_{U,V}) \subset [V - 8\sqrt{Vd}, V + 8\sqrt{Vd}].$$

Before deriving the product constant, the parameter branch and wedge give the non-circular coarse box

$$n \leq B \leq 3n, \qquad \frac{n}{2} \leq D \leq 2n, \qquad A \geq 8B, \qquad C \geq D + d, \qquad B, D \geq 256d.$$

Thus both Jacobi lower endpoints are positive and $B/D \leq 6$. The maximum-root product inequality is cited directly as MSS Theorem 1.6; applying it also to reciprocal polynomials supplies the lower endpoint. If u and v denote the deviations of the two factor roots from B and 1, respectively, then

$$\begin{aligned} |(B+u)(1+v) - B| &\leq 8\sqrt{Bd} + 8B\sqrt{d/D} + 64\sqrt{Bd}\sqrt{d/D} \\ &\leq (12 + 8\sqrt{6})\sqrt{Bd}. \end{aligned}$$

The resulting comparison roots satisfy

$$|y - B| \leq C_{\mathrm{loc}}\sqrt{Bd}, \qquad C_{\mathrm{loc}} = 12 + 8\sqrt{6}. \tag{6}$$

Critical points satisfy the same interval by interlacing. The elementary bound $\sqrt{6} < 5/2$ gives $C_{\mathrm{loc}} < 32$ and the exact rational sufficient choice $K_0 = 256 \cdot 32^2 = 262144$ for the strengthened box.

This argument does not use Holland's assembled Lemma 7.3: its hypothesis on $(C - D)/D$ fails on the present branch. It also corrects a theorem-number seam: the published MSS largest-root product inequality is Theorem 1.6.

9 The direct critical-point recurrence

For the mth derivative, shift the hypergeometric parameters to $(m-d, A+m, C+m; B+m, D+m)$. The coefficient ratio proves directly

$$\vartheta(\vartheta + B + m - 1)(\vartheta + D + m - 1)g_m = \lambda y(\vartheta + m - d)(\vartheta + A + m)(\vartheta + C + m)g_m,$$

Here

$$\vartheta = y\, \mathrm{d}/\mathrm{d}y, \qquad \lambda = D/(AC).$$

Expanding the Euler operators gives, for $T_k = y^k p_F^{(k)}(y)/p_F(y)$,

$$P_{3,m}T_{m+3} + P_{2,m}T_{m+2} + P_{1,m}T_{m+1} + P_{0,m}T_m = 0.$$

For a readable display of all four exact coefficients, set

$$\begin{aligned} a &= 1 - y/A, & b_m &= B + m - y + (d - 1 - 2m)y/A, \\ c_m &= (d - m)(1 + m/A)y, & \epsilon_p &= (C - D)/C, \\ \beta_m &= A(2m + 1 - d) - d(2m + 1) + 3m^2 + 3m + 1, & \Gamma_m &= m(m - d)(m + A). \end{aligned}$$

Then

$$\begin{aligned}
P_{3,m} &= a + \epsilon_p \frac{y}{A} = \frac{AC - Dy}{AC},\\
P_{2,m} &= b_{m+1} + (D+m)a + \epsilon_p \frac{y}{A}(A - d + 3 + 3m),\\
P_{1,m} &= c_{m+1} + (D+m)b_m + \epsilon_p \frac{y}{A}\beta_m,\\
P_{0,m} &= (D+m)c_m + \epsilon_p \frac{y}{A}\Gamma_m = \frac{Dy(A+m)(C+m)(d-m)}{AC}.
\end{aligned}$$

The expanded rational forms are also checked by Lean; the shifted hypergeometric ODE and these decompositions are checked by the exact symbolic producer.

At a critical point, $T_0 = 1$ and $T_1 = 0$. Localization (6), the wedge, and positivity of the non-perturbative $\epsilon_p = (C - D)/C$ term yield, uniformly for $0 \le m \le d - 2$,

$$P_{2,m} \ge n/8, \quad |P_{3,m}| \le 2, \quad |P_{1,m}| \ll n\sqrt{Bd} + nd, \quad 0 \le P_{0,m} \ll n^2 d.$$

A maximum argument in the recurrence then gives the critical-point derivative radius

$$\max_{1\le k\le d} \left| \frac{y^k p_F^{(k)}(y)}{p_F(y)} \right|^{1/k} \le K_r \sqrt{Bd} \tag{7}$$

with an absolute K_r; separately $T_0 = 1$. This is the decisive replacement for a perturbative recurrence that is unavailable when $\epsilon_p \to 1/2$.

10 Sixth-order residual

Let $E_F(z)$ be the logarithm of the exact xi/model coefficient ratio along the positive branch. Six differentiations give

$$\begin{aligned}
E_F^{(6)}(z) = h^{(6)}(n+z) &+ \psi^{(5)}(B+z) - \psi^{(5)}(n+1/2+z) - \psi^{(5)}(A+z)\\
&+ \psi^{(5)}(D+z) - \psi^{(5)}(C+z).
\end{aligned}$$

Pair the $B - C$ and $D - (n + 1/2)$ terms. On the thickened interpolation domain

$$\Omega = \{z : \operatorname{dist}(z, [0, d]) \le 2\rho\}, \qquad \rho = K_r\sqrt{Bd},$$

the wedge gives $d + 2\rho \le \eta n$ for a fixed small η. Hence $n + \Omega$ remains in the nonvanishing coefficient sector, while every straight polygamma segment stays in $\Re z \ge cn$. The absolutely convergent polygamma series and Section 5 imply

$$\sup_{z\in\Omega} |E_F^{(6)}(z)| \le \frac{C}{n^5 \log(n+2)}. \tag{8}$$

The complex Hermite–Genocchi formula is obtained by iterating the complex line-segment FTC over the standard six-simplex. Its mass is $1/6! = 1/720$. Since $E_F(0) = \cdots = E_F(5) = 0$, Newton interpolation and (8) give, for the needed z,

$$|E_F(z)| \le \frac{C}{720 n^5 \log(n+2)} \prod_{j=0}^{5} |z - j|. \tag{9}$$

The wedge implies $\rho \geq d$ after increasing its absolute constant. For $z \in \Omega$ and $0 \leq j \leq 5$, projection to $[0, d]$ then gives $|z - j| \leq 3\rho$. Since $B \asymp n$,

$$\rho^6 = K_r^6 (Bd)^3 \ll n^3 d^3,$$

and (9) yields the headline exponent calculation

$$\sup_{z \in \Omega} |E_F(z)| \ll \frac{\rho^6}{n^5 \log(n+2)} \ll \frac{d^3}{n^2 \log(n+2)} \leq \frac{C'}{K}. \tag{10}$$

Choose the absolute wedge constant so that the last quantity is small enough that, for $c_F(z) = e^{E_F(z)}$, $\sup_\Omega |c_F - 1| \leq 1$. Exponentiation is legal on the same chosen logarithm branch. At the integer nodes the exact coefficient ratios are positive real numbers, and the six matches give $c_F(0) = \cdots = c_F(5) = 1$.

11 Sign transfer and proof of the main theorem

Lemma 11.1 (Finite multiplier stability)**.** *Let $d \geq 5$, and let $p(y) = \sum_{j=0}^d p_j y^j$ be real with d simple positive zeros and $p(0) \neq 0$. Put $\Omega_r = \{z : \operatorname{dist}(z, [0, d]) \leq 2r\}$, and let c be holomorphic on a neighborhood of Ω_r. Suppose that $c(0), \ldots, c(d)$ are real, $c(0) = \cdots = c(4) = 1$, $c(d) > 0$, $\sup_{\Omega_r} |c - 1| \leq \epsilon < 16$, and, at every critical point y of p,*

$$\left| \frac{y^k p^{(k)}(y)}{p(y)} \right| \leq r^k, \qquad 0 \leq k \leq d.$$

Then $P(y) = \sum_{j=0}^d p_j c(j) y^j$ has d distinct positive real zeros.

Proof. Newton interpolation and termwise differentiation give

$$\frac{P(y)}{p(y)} = \sum_{k=0}^d \frac{\Delta^k c(0)}{k!} \frac{y^k p^{(k)}(y)}{p(y)}.$$

Cauchy's estimate on the radius-$2r$ discs in Ω_r gives $|\Delta^k c(0)|/k! \leq \epsilon/(2r)^k$. The five exact matches make the terms $1 \leq k \leq 4$ vanish, so at every critical point

$$\left| \frac{P(y)}{p(y)} - 1 \right| \leq \epsilon \sum_{k=5}^d 2^{-k} < \epsilon/16 < 1.$$

Thus P and p have the same signs at all critical points, at zero, and for large positive y. Interlacing and the intermediate-value theorem put one simple sign-changing zero of P in each of the d intervening intervals. □

Proof of Theorem 1.1. The comparison polynomial has d simple positive roots. Between them lie $d - 1$ critical points, and the two exterior sign tests complete the alternating pattern. For $d \leq 5$, the six coefficient matches already cover the whole polynomial. For $d \geq 6$, apply Lemma 11.1 with $p = p_F$, $c = c_F$, and $r = \rho$. Equations (7), (10), and the positivity of the integer coefficient ratios verify every hypothesis with $\epsilon \leq 1 < 16$. The lemma therefore preserves every comparison sign, so the intermediate-value theorem supplies d distinct positive roots of the transformed actual polynomial.

The positive parameter branch has a positive scale and nonzero normalization. Undoing $y = -SX$, with $S > 0$, carries those roots to d distinct negative roots of $\mathcal{J}^{d,n}$. This proves Theorem 1.1. The low finite set below the eventual analytic threshold is absorbed by increasing the existential constant K; when the wedge is empty there is nothing to check. □

12 Formal verification and trust boundary

The Lean project contains no axiom asserting the xi asymptotic or the headline result. It checks the complete concrete T1–T5 chain: the xi/theta/Mellin identity, quantitative saddle branch, legal leading contour, infinite higher-mode suppression, Gamma/Stirling assembly, sectorial coefficient theorem, and Cauchy transport through order six. It also checks the rational leading system and Jacobian, quotient-to-six-coefficient adapter, cube calculus, exact xi branch and six matches, direct recurrence, multiplier sign transfer, concrete Jensen definition, and positive-to-negative scaling.

The complex Hermite–Genocchi modules prove complex line-segment FTC, six repeated FTC steps, the exact Newton identity, convex-hull localization, the six-simplex mass 1/720, and the local remainder bound. The approved Phase-30 endpoint constructs the concrete xi multiplier, proves the first six node values and its strict unit bound, instantiates the complete Rolle-point interval certificate, and connects sign transfer to the actual transformed Jensen polynomial. Classical Jacobi root/matrix facts, MMP log-mesh preservation, and the MSS product-root inequality remain narrow typed literature inputs. These limitations are proof-surface disclosures, not hidden axioms.

The literal T5 statements have been isolated in a locally verified Phase-29 Palomar submission candidate. Official Comparator and NanoDa replay remains pending, so no official Palomar result is claimed.

The verifier scans for `sorry`, `admit`, custom axioms, and unsafe escape hatches, builds the headline target with the pinned toolchain, and audits selected theorem axioms. Numerical contour and Mellin checks are regression evidence only.

13 Independent-CAS Mathematica reconstruction

A user-executed Mathematica 15.0.1 reconstruction returned exact matches for M1–M4. The user entered and executed cells supplied interactively by Codex; the notebook imported no repository JSON, Python, SymPy, Lean-generated expression, or frozen numerator. Thus the run breaks the SymPy-versus-Mathematica CAS common mode, while remaining AI-assisted evidence rather than human or peer review.

The exact terminal results were: the implicit saddle derivative identity and post-chain constants $(2, -2, 4, -12, 48)$; the shifted ${}_3F_2$ coefficient recurrence, four zero coefficient differences, and complete derivative-order tests at degrees $5, 8, 11$; the positive solution $(3, 2, 16/3, 1/3)$, determinant $-1/144$, and inverse norm $304/3$; and the 82-term degree-13 H_6 numerator with coefficientwise majorant

$$\frac{6422139805764931584036533551104}{702576099728137594188684005} < 10^4.$$

All mathematical outputs are exact and contain no machine-real values. The notebook, plain-text result ledger, visual PDF, and user checksum ledger are frozen in the Phase-24 evidence directory. The canonical result-ledger SHA-256 is `1faea5fcb35b504b5d0aad9391999a99e530373dce36addb496eabb133fb04cc`.

14 Availability and review statement

The Version 1.0 public-release set is designed to contain this source and rendered PDF, all proof notes needed to expand the compressed arguments, the Lean source and lockfiles, exact symbolic and interval certificates, verifier scripts, serial logs, primary-source hashes, and AI-review disclosures. Public posting has been authorized, but the absence of human mathematical or peer review remains a material limitation and is disclosed on every circulation surface. The public version should be cited by its version-specific DOI and checksum once registered; corrections should appear as new versions rather than silent file replacement.

References

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